SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I THIRU KUMARAN S / 192521149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S.

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.
   Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
   Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.
- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.

- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

SET- C

1. Cloud Migration Strategy

introduction:

cloud migration means moving applications, databases, and data from a local data center to the cloud. A manufacturing company can use cloud migration to improve performance, security, and availability.

1. cloud migration strategy

The company can follow these steps:

* Assessment
* planning
* Backup
* Migration
* Testing
* Monitoring

2. Type of cloud migration:

* Rehost
* Refactor
* Re platform
* Repurchase
* Retire
* Retain.

3. Performance validation

After migration, the Company should:

* check application response time.
* Test database performance.
* check data accuracy
* Test the application with multiple users.
* Monitor CPU, memory, and network usage.

4. Reducing Risks and downtime:

The Company should take regular backups and migrate applications in stages. It should test applications before making them available to all users. Data replication helps keep data updated.

Security can be improved by using encryption and access control.

2. Smart City IoT Architecture

Introduction:

IoT stands for internet of Things. It connects physical devices to the internet to collect and share data

Smart cities use IoT sensors to monitor traffic, air quality, energy consumption, and Public Safety.

cloud-based IoT Architecture

* IoT Sensors
* Network
* Message Broker
* Stream processing
* cloud storage
* Analytics
* Dashboard.

2. Integration of message Brokers, Stream processing, and cloud Storage

* message brokers receive data from different sensors and it to correct services.
* stream processing analyzes data immediately.
* All these components work together to provide real time information.

3. Handling Large Amounts of Data.

* use distributed storage to store large amounts of Data
* Remove unnecessary data.
* use auto-scaling when the no of sensors increases.

4. Scalability, Reliability, and Security

* Scalability can be achieved by adding cloud resources.
* Reliability can be improved using backups
* Security can be encryption protects information.

3 Business intelligence Architecture:

Introduction:

Business intelligence (BI) is the process of collecting & analyzing data

1. Cloud-based BI Architecture
   * Data Sources:
   * Data Ingestion
   * ETL process
   * cloud Data warehouse
   * Analytics
   * visualization tools
   *
   * Dashboard.

2. Visualization Tools and Data warehouses:
   * Data is collected from different sources
   * The processed data is stored in a cloud data warehouse.
   * executives can easily understand sales performance.
   * Reports can be updated regularly.

3. improving Dashboard performance:
   * using Optimized database queries
   * Storing frequently used result in a cache
   * using summarized data
   * Avoiding unnecessary data
   * using Scalable cloud resources.

Cloud Monitoring and performance optimization:

1. Cloud monitoring Solution:

* Monitoring Tools
* Metrics
* Logs
* Tracing
* Load Balancing
* Auto-Scaling
* Alerts.

2. Role of monitoring Tools, Metrics, Logs, & Tracing

* Monitoring tools show the overall performance of the system.
* Metrics helps identify high cpu usage
* Logs record errors & activities
* Tracing shows how a request moves

3. Resource optimization and Capacity

* Using auto-scaling during peak hours.
* optimizing database queries.
* using caching to reduce repeated work.

4. Improving Availability & Reducing costs.

The Company should deploy servers across multiple availabity zones.

Regular backups help protect important data.

The Company should use cloud resources according to demand and remove unnecessary services.

5. Data Lifecycle Management.

Introduction:

Data lifecycle management is the process of managing data from creating to deletion.

1. Cloud-based Data lifecycle:
   * Collection
   * Classification
   * Storage
   * Usage
   * Archiving
   * Deletion.
2. Data classification & Retention
   * classification
   * Retention
   * Archiving
   * lifecycle policies
3. Governance, Auditing, & Compliance
   * Auditing records user activities.
   * Encryption protects sensitive data
   * Access control prevents unauthorized access.
4. Cost optimization & Security
   * Take regular backups
   * use encryption and access control
   * Monitor storage usage & costs.

Conclusion:

cloud data lifecycle management helps financial institutions protect customer data, follow regulations & reduce storage costs.